

Reg. No:

Name:

APJ ABDULKALAM TECHNOLOGICAL UNIVERSITY
SECOND SEMESTER M.TECH. DEGREE EXAMINATION SEPTEMBER 2023

Course Code: 222ECS016

Course Name: KNOWLEDGE BASED SYSTEM DESIGN

Max. Marks: 60

Duration: 2.5 hours

Branch : Computer Science and Engineering

PART A

(Answer all questions in PART A. Each question carries 5 marks)

Q. No.		Marks
1	Discuss the different characteristics of knowledge.	(5)
2	With an example, illustrate the working of a commercial expert system.	(5)
3	Explain briefly the objectives of developing a KBS.	(5)
4	What is the need of a distinct development model for a KBD? How it differs from other learning models?	(5)
5	Write a short note on modelling of uncertain reasoning.	(5)

PART B

(Answer any five full questions in PART B. Each full question carries 7 marks)

6	a. Explain the basic structure of knowledge-based system.	(3)
	b. Describe the working of knowledge base and inference Engine.	(4)
7	a. What are the tools that can be used to build expert systems?	(3)
	b. Explain a search space approach for solving tic tac toe game.	(4)
8	a. Give an outline on rule-based systems.	(3)
	b. Using appropriate examples, explain the different categories of KBS.	(4)
9	a. Illustrate the working of inference engine in a KBD.	(3)
	b. Elaborate on the various difficulties to be faced while developing a KBS	(4)
10	Explain the different strategies for developing relationship with experts and dealing with multiple experts while implementing a knowledge-based system.	(7)

- 11 Illustrate knowledge-based system development model. Explain the basic activities of knowledge-based systems development. (7)
- 12 a. With suitable examples, Point out the differences between case-based reasoning and rule-based reasoning. (3)
- b. With an example, explain machine learning based problem-solving approach, its learning evaluation and testing. (4)